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Group 9

Life Cycle Analysis of Wireless Mouse: Environmental Impact Assessment

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Executive Summary

Life Cycle Assessment (LCA) is very important in today's world of sustainable development, as it can comprehensively analyze the environmental impacts of products at all stages, from raw material extraction, production, use to final disposal, and provide scientific data and quantitative indicators to help companies or governments make the right decisions when formulating relevant policies. Considering that, this study was conducted to understand the environmental impact of a computer mouse which is widely used as a mandatory electronic device in the current world with an exponential global demand. Therefore, the objective of this study was to conduct a comprehensive LCA based on a 'cradle-to-grave' approach, covering the entire cycle of raw material extraction, production, use and eventual disposal of the end-of-life mouse. To ensure the comprehensiveness of the project, the Open LCA 2.2 tool and the Eco Invent 3.7 database for data processing and evaluation based on the ISO 14040 standard were employed. During the study the mouse was disassembled and analyzed carefully to quantify each part of the mouse (e.g. casing, internal wiring, structural components, and circuit boards), constructed a detailed life cycle inventory (LCI) by collecting data related to raw material use, power consumption, and transport routes, and assessed the environmental impacts using the ReCiPe 2016 midpoint methodology.

The study reveals that the material phase of a computer mouse contributes the most to global warming potential (64.2% of total 1.75 kg CO₂ eq.), fossil resource scarcity, and freshwater ecotoxicity due to energy-intensive raw material extraction and processing. The use phase, driven by battery consumption, accounts for approximately 28% of GWP and significantly impacts categories such as human carcinogenic toxicity and marine eutrophication. Hotspot analysis identifies key contributors to environmental impact, including the printed wiring board (PWB) which alone accounts for nearly 0.8 kg CO₂ eq, as well as batteries and ABS plastic. Sensitivity analysis shows that rail transport in China can reduce greenhouse gas emissions by up to 33% in certain categories compared to road transport in India. The findings emphasize the importance of material selection and transportation modes in minimizing environmental impacts.

1.0 Introduction

The global market for personal computers, including desktops, laptops, and peripherals like mice, is projected to be valued at USD 204.07 billion in 2024, growing to USD 315.42 billion by 2029 with a compound annual growth rate (CAGR) of 9.10% (Mordor Intelligence, 2024). This growth reflects rising consumer and business demand for PCs and accessories like computer mice. Understanding the environmental impacts of these devices is essential for industry regulation. However, few studies quantify these impacts. This project aims to assess the environmental effects of a computer mouse using a cradle-to-grave Life Cycle Analysis (LCA) approach.

2.0 Methodology

LCA, guided by the ISO 14040 and ISO 14044 standards, is a comprehensive and widely recognized tool for assessing the environmental impacts of products and processes throughout their entire life cycle, from raw material extraction to disposal (Finkbeiner et al., 2006). A traditional LCA has four stages, such as (i) Goal and scope definition, (ii) Life Cycle Inventory (LCI), (iii) Life Cycle Impact Assessment (LCIA), and (iv) Interpretation (Muralikrishna and Manickam, 2017). This project employs the LCA tool using ISO 14040 methodology and covering the entire lifecycle from raw material extraction through production, usage, and end-of-life disposal of a computer mouse. For that, the Open LCA 2.2 and the EcoInvent 3.7 database were employed in this project.

2.1 Goal & Scope Definition

The studied computer mouse model is from Logitech, a Swiss brand company (Logitech, 2024). As per the Logitech website, it was identified that the studied mouse was manufactured in China and distributed in Australia. Besides, this device has several components made with various varied materials. Therefore, to quantify the environmental impacts of this product, it is essential to construct a system boundary and set a functional unit extended till the disposal phase on which the impacts will be calculated.

2.1.1 System Boundary

This project adopts a cradle-to-grave approach, as illustrated in **Fig. 1**. Consequently, the scope of the study extends from the extraction of raw materials to the disposal of the product after its maximum use. At each stage, materials, energy, and, in some cases, water enter the system as inputs. As a result, emissions to air, water, and soil occur alongside the desired product. Additionally, throughout every stage of its life cycle, materials and components are transported from one location to another. Therefore, beyond the raw material extraction and manufacturing/assembly processes, transportation is a significant aspect to consider. Although various transportation modes are integrated throughout the entire life cycle, it is treated as a distinct phase in this project.

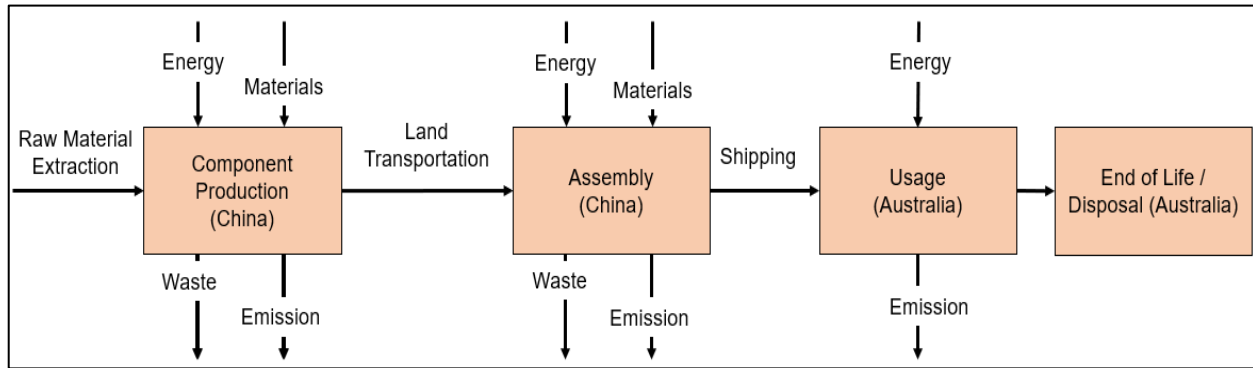


Fig. 1 System boundary of the LCA study

2.1.2 Functional Unit

In a typical office environment, a computer mouse is used daily for an 8-hour workday, with an average of around 400 clicks per day. With proper maintenance, a computer mouse can have an average lifespan of 3 years (MeeTion, 2024). Therefore, the functional unit defined for this project is “a wireless Bluetooth mouse to perform average 400 clicks daily over a period of 3 years.”

2.2 Life Cycle Inventory (LCI) and Analysis

The LCI for the computer mouse involves the systematic collection of data on the inputs and outputs associated with each stage of its life cycle, from raw material extraction, manufacturing, and assembly to its use, transportation, and eventual disposal. This data includes the consumption of resources like energy and materials, providing a detailed foundation for assessing the environmental impacts of the computer mouse.

2.2.1 Data collection and assumptions

2.2.1.1 Components and the measurements

To analyze the materials and assembly processes (foreground data), the mouse was dismantled in the TCS lab, weighed, and its materials were characterized based on their properties and polymer types.

The Bluetooth mouse is made from various materials selected for functionality, durability, and efficiency (see Fig. 2(b)). The outer casing, made from ABS (Acrylonitrile Butadiene Styrene) plastic, weighs 39.25 grams, chosen for its impact resistance and ease of molding, protecting internal components (Rutkowski & Levin, 2021). The internal wiring, weighing 0.07 grams, is copper, valued for its electrical conductivity (Copper Development Association, 2023). HDPE (High-Density Polyethylene) makes up structural components, weighing 0.52 grams, known for its strength and lightweight properties (American Chemistry Council, 2022). Stainless steel parts like the scroll wheel, weighing 1.76 grams, are selected for durability and corrosion resistance (International Stainless-Steel Forum, 2024). The circuit board weighs 5.5 grams, made from

fiberglass and metals, while the LED optical sensor, weighing 0.22 grams, uses gallium and silicon for precise tracking (Schubert, 2023). Table 1 summarizes the total material mass.

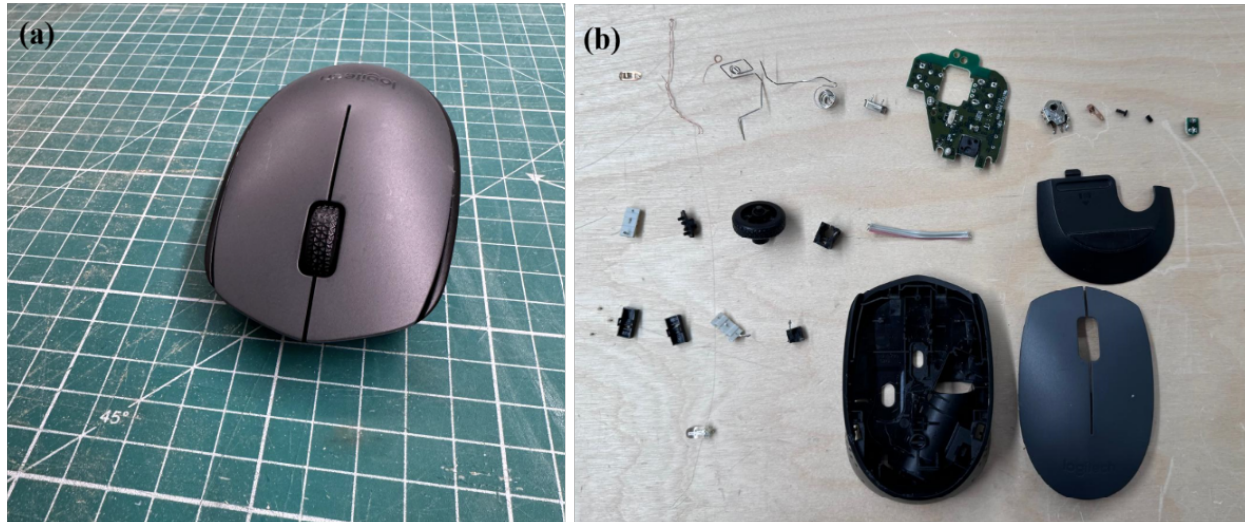


Fig. 2 Studied computer mouse (a) before the dismantling process, (b) dismantled components

Moreover, the circuit board, weighing 5.5 grams, is composed of fiberglass, copper, tin, and rare metals, crucial for electronic functionality (IPC, 2024). Additionally, the LED optical sensor weighs 0.22 grams and uses materials like gallium and silicon for precise tracking (Schubert, 2023). Each component is selected to maximize performance, durability, and user experience. The total mass of materials and utilities required for the assembly process is summarized in Table 1.

Table 1: Material and utilities for Assembly

Material/Process	Mass/Consumption
ABS Plastic Molding	39.25 g
Copper Wire Making	0.07 g
HDPE Molding	0.52 g
Stainless Steel Wire Making	1.76 g
Electricity (assembly)	0.06 kWh
Water	0.01 kg

2.2.1.2 Production Location

Logitech company has a factory located in the Suzhou area of China (Zhong Nan, 2024). The packaging of the computer mouse indicates that the item is made in China but does not specify the region. Therefore, assumption is made for the Logitech computer mouse is made in Suzhou region, near to Shanghai.

2.2.1.3 Transportation

The transport routes and methods for materials and final products in this study involve multiple stages and modes of transportation, detailed in Appendix Table A.1 and visually shown in Fig. A.1. Raw materials like steel wire, copper wire, cardboard, and plastic are produced in Wuhan, China, and transported 840.5 kilometers by heavy trucks to Shanghai for further processing or assembly (Rodrigue, 2020), followed by shipment to the Port of Melbourne, Australia, covering 7,981 kilometers by sea (EPA, 2023). After arriving in Australia, the product undergoes additional transport stages, including heavy truck delivery to a regional warehouse, light truck distribution to retailers, and small car delivery to end customers, each stage contributing to the overall environmental impact through fuel consumption and emissions (Victorian Auditor-General's Office, 2022; Steg et al., 2014).

2.2.2 Life cycle inventory development

Each component of the Bluetooth mouse was weighed in grams, and their materials were identified through research. The inventory analysis identified key materials such as plastics, metals, and electronic components, each chosen for their specific properties and functions. The detailed LCI chart is in **Table 2**.

Table 2: Life cycle inventory of the computer mouse

Phase	Input	Amount	Unit	Ecoinvent background data
Material	ABS plastic	39.25	g	market for acrylonitrile-butadiene-styrene copolymer acrylonitrile-butadiene-styrene copolymer Cutoff, S - GLO
	Copper	0.07	g	market for copper, cathode copper, cathode Cutoff, S - GLO
	HDPE	0.52	g	market for polyethylene, high density, granulate polyethylene, high density, granulate Cutoff, S - GLO
	Stainless steel	1.76	g	market for iron-nickel-chromium alloy iron-nickel-chromium alloy Cutoff, S - GLO
	Cable with copper wire+ PVC	0.25	g	market for cable, unspecified cable, unspecified Cutoff, S - GLO
	Printed wiring Board (PWB) (5.5 gram)	24	cm ²	market for printed wiring board, for surface mounting, Pb containing surface printed wiring board, for surface mounting, Pb containing surface Cutoff, S - GLO
	LED Light (IR movement detector)	0.22	g	market for light emitting diode light emitting diode Cutoff, S - GLO
	Printed Cardboard	6	g	market for folding boxboard carton folding boxboard carton Cutoff, S - RoW
	Plastic packaging	0.69	g	market for packaging film, low density polyethylene packaging film, low density polyethylene Cutoff, S - GLO

Phase	Input	Amount	Unit	Ecoinvent background data
Manufacturing/ Assembling	ABS plastic Moulding	39.25	g	market for injection moulding injection moulding Cutoff, S - GLO
	Copper wire making	0.07	g	market for wire drawing, copper wire drawing, copper Cutoff, S - GLO
	HDPE Moulding	0.52	g	market for injection moulding injection moulding Cutoff, S - GLO
	Stainless steel wire making	1.76	g	market for wire drawing, steel wire drawing, steel Cutoff, S - GLO
	Electricity (assembly)	0.06	kWh	electricity voltage transformation from high to medium voltage electricity, medium voltage APOS, S - CN-CSG
	Water	0.01	KG	market for tap water tap water Cutoff, S - RoW
	AA battery (1.5 V)-3 pcs	69.0	g	market for battery, NaCl battery, NaCl Cutoff, S - GLO
Use	Consumer electronics waste management	141.76	g	treatment of waste plastic, consumer electronics, dismantling, sorting and cleaning, formal sector waste plastic, consumer electronics, sorted Cutoff, S - IN
	Landfill (consumer electronic waste) (70%)	99.23	g	treatment of waste plastic, consumer electronics, sanitary landfill, wet infiltration class (500mm) waste plastic, consumer electronics Cutoff, S - GLO
EoL	Steel wire _Wuhan to shanghai	1.48	KG-KM	market group for transport, freight, lorry, unspecified transport, freight, lorry, unspecified Cutoff, S - GLO
	Copper wire _Wuhan to shanghai	0.06	KG-KM	market group for transport, freight, lorry, unspecified transport, freight, lorry, unspecified Cutoff, S - GLO
	Cardboard+Plastic _Shnghai-Factory	0.33	KG-KM	market group for transport, freight, lorry, unspecified transport, freight, lorry, unspecified Cutoff, S - GLO
	Final product _Factory - Port	3.64	KG-KM	market group for transport, freight, lorry, unspecified transport, freight, lorry, unspecified Cutoff, S - GLO
	Final product _Shanghai-Melbourne	580.70	KG-KM	market for transport, freight, sea, container ship transport, freight, sea, container ship Cutoff, S - GLO
	Final product _Port - stores	4.37	KG-KM	market for transport, freight, light commercial vehicle, EURO1 transport, freight, light commercial vehicle, EURO1 Cutoff, S - ZA
	Final product _Store-house	0.36	KG-KM	market for transport, freight, light commercial vehicle, EURO1 transport, freight, light commercial vehicle, EURO1 Cutoff, S - ZA
	Final product _Store-house	0.35	KG-KM	market for transport, freight, light commercial vehicle, EURO1 transport, freight, light commercial vehicle, EURO1 Cutoff, S - ZA

2.3 Life cycle impact assessment (LCIA)

For the LCIA, this study used the ReCiPe 2016 midpoint impact assessment method, which converts LCI data into environmental impacts across various categories like climate change, ozone depletion, and resource depletion (Huijbregts et al., 2016). **Figure 3** shows the environmental impacts across life cycle stages, with the Material and Use phases being the most significant. The Material phase contributes 15% to 65%, and the Use phase 22% to 85%, due to energy-intensive material extraction and battery (3 AA batteries) use. Although the End of Life (EoL) phase has a minor impact, it significantly affects Marine Eutrophication (~55%).

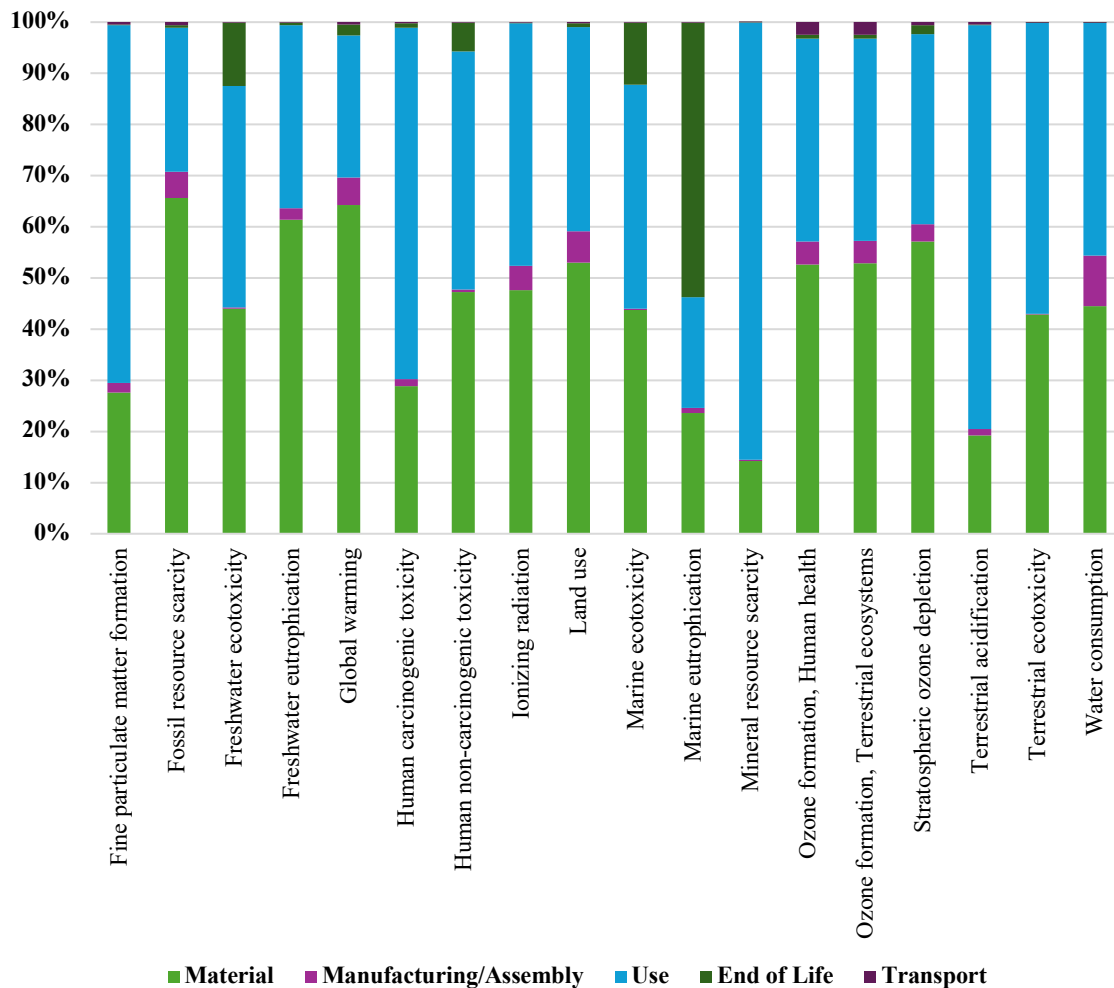


Fig. 3 Life cycle Midpoint impacts of the studied computer mouse

Table A.2 outlines the environmental impacts across different life cycle stages of the mouse. The global warming potential amounts to 1.75 kg CO₂ eq, with the material phase contributing the largest portion (~64.2%) due to energy-intensive raw material extraction and processing. The use phase follows with ~28%, attributed to energy consumption during product usage. Manufacturing

adds 0.094 kg CO₂ eq, while transport and end-of-life stages contribute minor portions (0.05% and 2%, respectively) For fossil resource scarcity, the material phase contributes the most (~66.25%) with 0.48 kg oil eq, followed by the use phase (~28%). Freshwater ecotoxicity impacts are significant in both the material and use phases, totaling 0.7233 kg 1,4-DCB. Human carcinogenic toxicity is primarily driven by the use phase (0.2374 kg 1,4-DCB), contributing ~69%, while marine eutrophication and terrestrial ecotoxicity also see significant impact from the use phase (0.0000524 kg N eq and 25.39 kg 1,4-DCB, respectively).

2.4 Hotspot analysis

Since climate change due to global warming is a major concern in the current world, this employs the GWP from the midpoint impact categories to analyze the hotspot in the life cycle of the studied computer mouse. However, the hotspots can be different in other midpoint impact categories.

In the hotspot analysis, shown in **Fig. 4**, the most significant contributors to GWP during the computer mouse's production process are identified. The LCA identifies three key hotspots contributing significantly to its environmental impact: the PWB, the battery, and the ABS plastic material. The PWB, accounting for nearly 0.8 kg CO₂ eq, has a high impact due to energy-intensive manufacturing processes and the use of materials like copper and fiberglass (Pecht, 2021; Liu et al., 2022). Batteries, particularly disposable types, contribute substantially to GWP due to the mining of metals and energy-intensive production processes (Dunn et al., 2020; Xu et al., 2021). ABS plastic, derived from petrochemicals, adds significantly to emissions due to its high energy production requirements (Song et al., 2020; Zheng et al., 2021).

Apart from these, electricity consumed in the assembling process, injection molding of both the ABS and HDPE combinedly in the manufacturing process, and LED light used as the pointer have significant contributions in the overall impact in this category (i.e., 2.4%, 2.9%, and 3.4%, respectively).

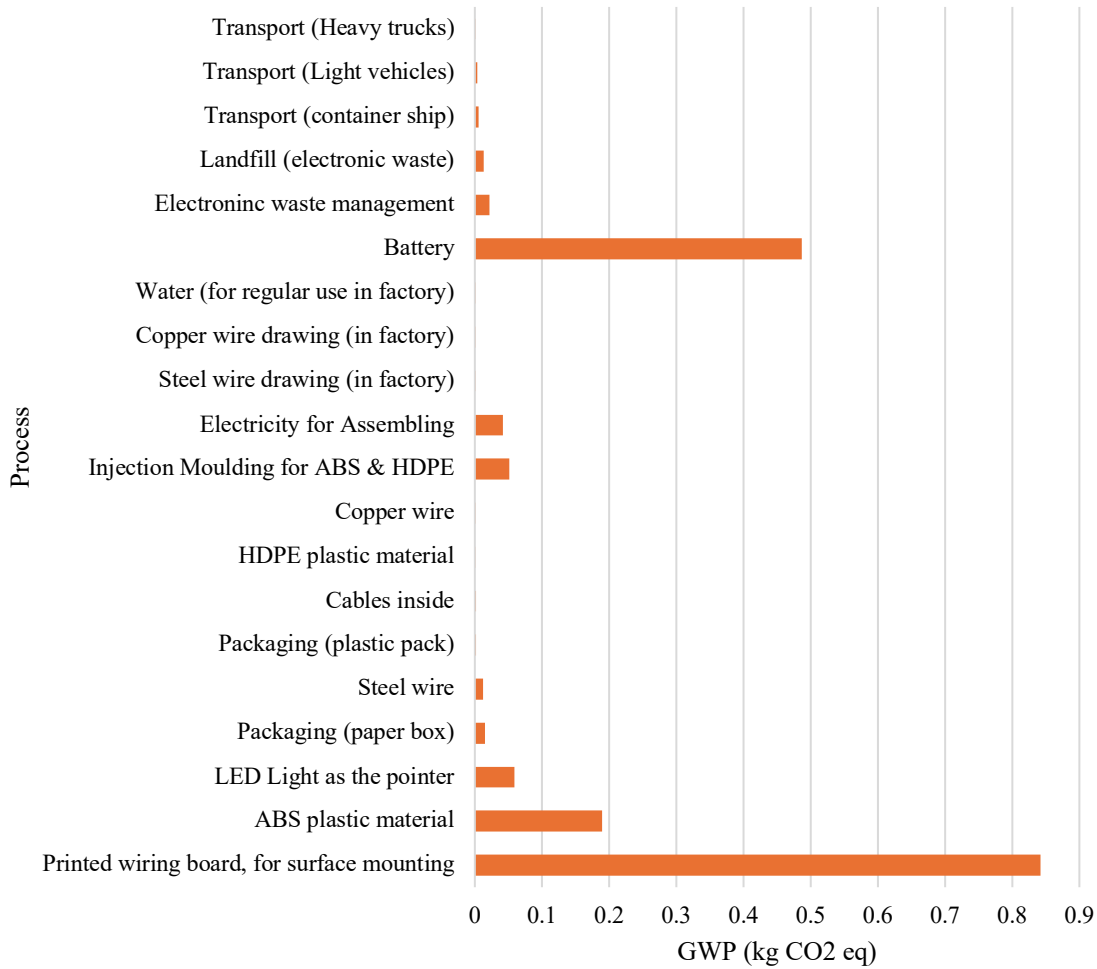


Fig. 4 Hotspot analysis in the life cycle of the computer mouse

2.6 Sensitivity analysis

The sensitivity analysis was conducted by comparing the transportation phase of the Logitech computer mouse across three different modes and regions. The baseline for this analysis is the use of road transport in China, which serves as the midpoint. The baseline is then compared with two other modes: road transport in India and rail transport in China (see **Table A.3**). The comparison results allow for a detailed examination of the environmental impacts based on the different transportation modes. Based on the comparison results in **Fig. 5**, a key analysis of the significant environmental impacts has been identified. In general, China's rail transport emerges as the most environmentally friendly option in multiple categories in compare with the other two modes. For instance, in terms of Terrestrial ecotoxicity and Land use, rail transport significantly reduces the negative effects compared to road transport in both China and India (maximum ~33% and ~18%, respectively).

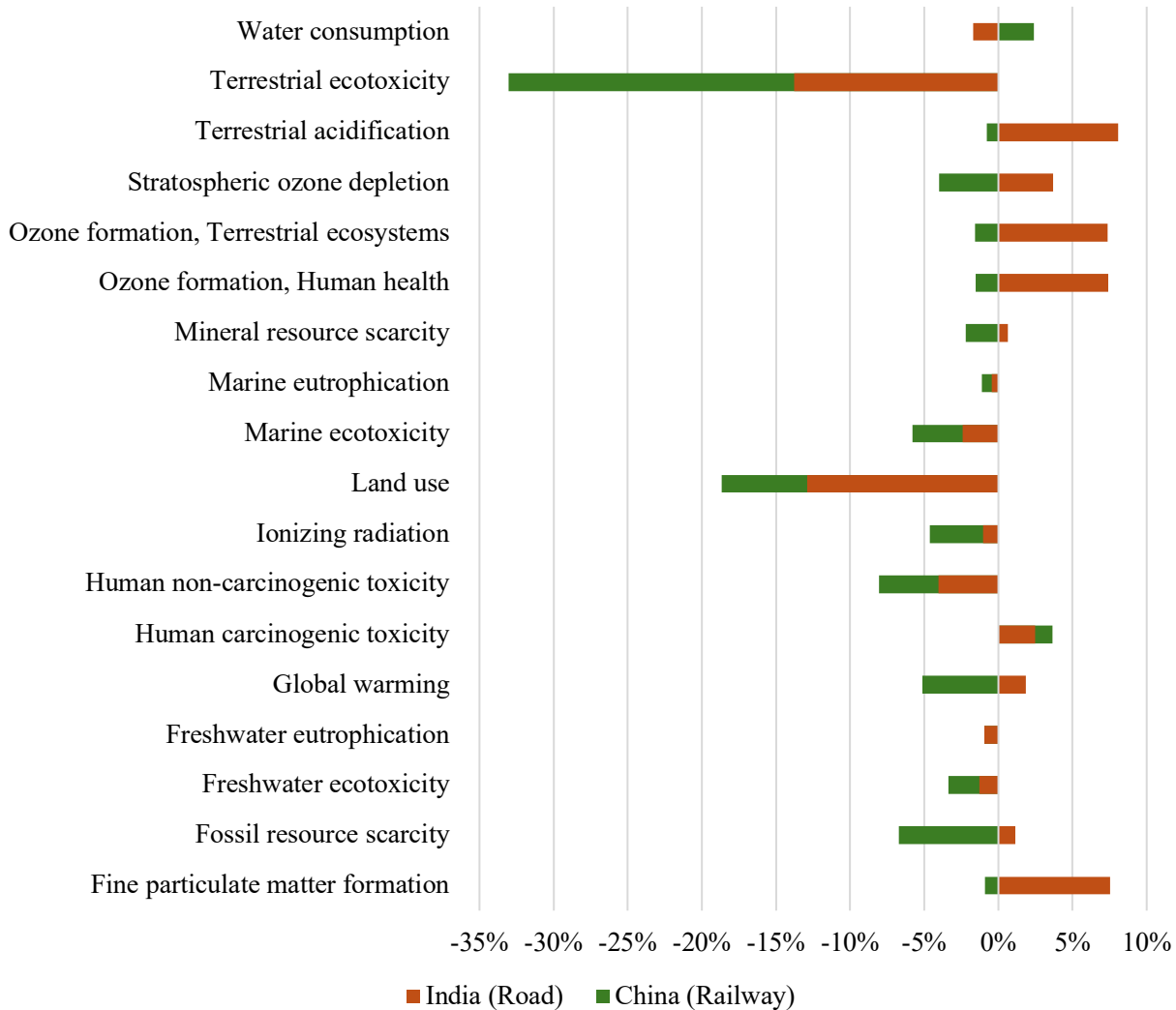


Fig. 5 Sensitivity analysis for the Transport phase

In the area of global warming, rail transport in China also shows its benefits, with lower greenhouse gas emissions compared to road transport, especially in India. Therefore, the impact may vary between -5% to +2%. In addition, a study by Li et al (2022) shows that the introduction of high-speed rail in China has significantly reduced carbon emissions and environmental impacts. Similar trends can be seen in the stratospheric ozone depletion, mineral resource scarcity, and fossil resource scarcity. Besides, in any of the transport modes, the human carcinogenic toxicity (maximum 4%) will be higher than the baseline. Contrary, in freshwater eutrophication, freshwater ecotoxicity, marine eutrophication categories, the overall impact of the mouse from the transport phase will be lesser than the baseline scenario (i.e., -1%, -4%, -1%, respectively).

However, the sensitivity analysis reveals that China's rail transport offers clear advantages in reducing environmental impacts across these critical categories, while India's road transport results in significantly higher negative impacts. This highlights the importance of considering transportation modes and regions when assessing environmental sustainability.

3.0 Discussion

The LCA of wireless mouse reveals that the manufacturing and use phases are the primary contributors to environmental impacts, particularly due to the production of PWB and ABS plastics. These processes are energy-intensive and emit hazardous substances, increasing global warming potential and resource depletion risks. Additionally, the use of disposable batteries significantly affects global warming and ozone formation. Hotspot analysis identifies PWBs, ABS plastics, and disposable batteries as the main environmental culprits, impacting freshwater ecotoxicity, eutrophication, and carbon emissions. Transport, especially long-distance shipping and trucking, also adds to CO₂ emissions.

Mitigation strategies include using renewable or recyclable materials instead of ABS plastics, optimizing manufacturing processes to reduce energy use, promoting rechargeable batteries to minimize battery waste, and shifting to rail transport to reduce emissions (Harper et al., 2019; Wu et al., 2020). Studies suggest bio-based and biodegradable materials can lower the carbon footprint, while cleaner production technologies can cut energy consumption. Comparisons with other LCA studies confirm similar findings, particularly regarding the environmental impact of PCBs and ABS plastics. This analysis highlights effective strategies for reducing the environmental footprint of wireless mice through material substitution, process efficiency, and transport optimization.

4.0 Conclusion

This project assessed the environmental impacts of wireless mice during the manufacturing, use and disposal phases through LCA. The findings show that the material and use phases have the greatest impact on almost all the categories. So, it is important to focus on these two phases and the components and materials used in these phases. The designers and brands, such as Logitech, can specific measures to reduce the impacts and environmental footprints. For instance, to reduce the environmental impact, this study can propose to replace traditional plastics with renewable materials or recycled plastics, optimize the manufacturing process to improve energy efficiency, promote battery recycling and prioritize the use of more environmentally friendly means such as rail transport. These measures will help to reduce the overall environmental impact of the process and drive it in a more sustainable direction, in line with eco-design principles.

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Appendix

Table A.1 Assumption of the transport distances in the life cycle of the computer mouse

Material/Product	Transport	Destination	Transport Method	Distance (KM)
Steel Wire Production	Wuhan to Shanghai	China	Heavy Truck	840.5
Copper Wire Production	Wuhan to Shanghai	China	Heavy Truck	840.5
Cardboard Production	Wuhan to Shanghai	China	Heavy Truck	840.5
Plastic Production	Wuhan to Shanghai	China	Heavy Truck	840.5
Mouse Production and Assembly	Shanghai to Port of Shanghai	China	Heavy Truck	50
Mouse (Final Product)	Port of Shanghai to Port of Melbourne	China to Australia	Ocean Going Ship	7981
Mouse (Final Product)	Port of Melbourne to Regional Warehouse	Australia	Heavy Truck	50
Mouse (Final Product)	Regional Warehouse to Retailer Store	Melbourne, Australia	Light Truck	20
Mouse (Final Product)	Retailer Store to Customer	Melbourne, Australia	Small Car	5

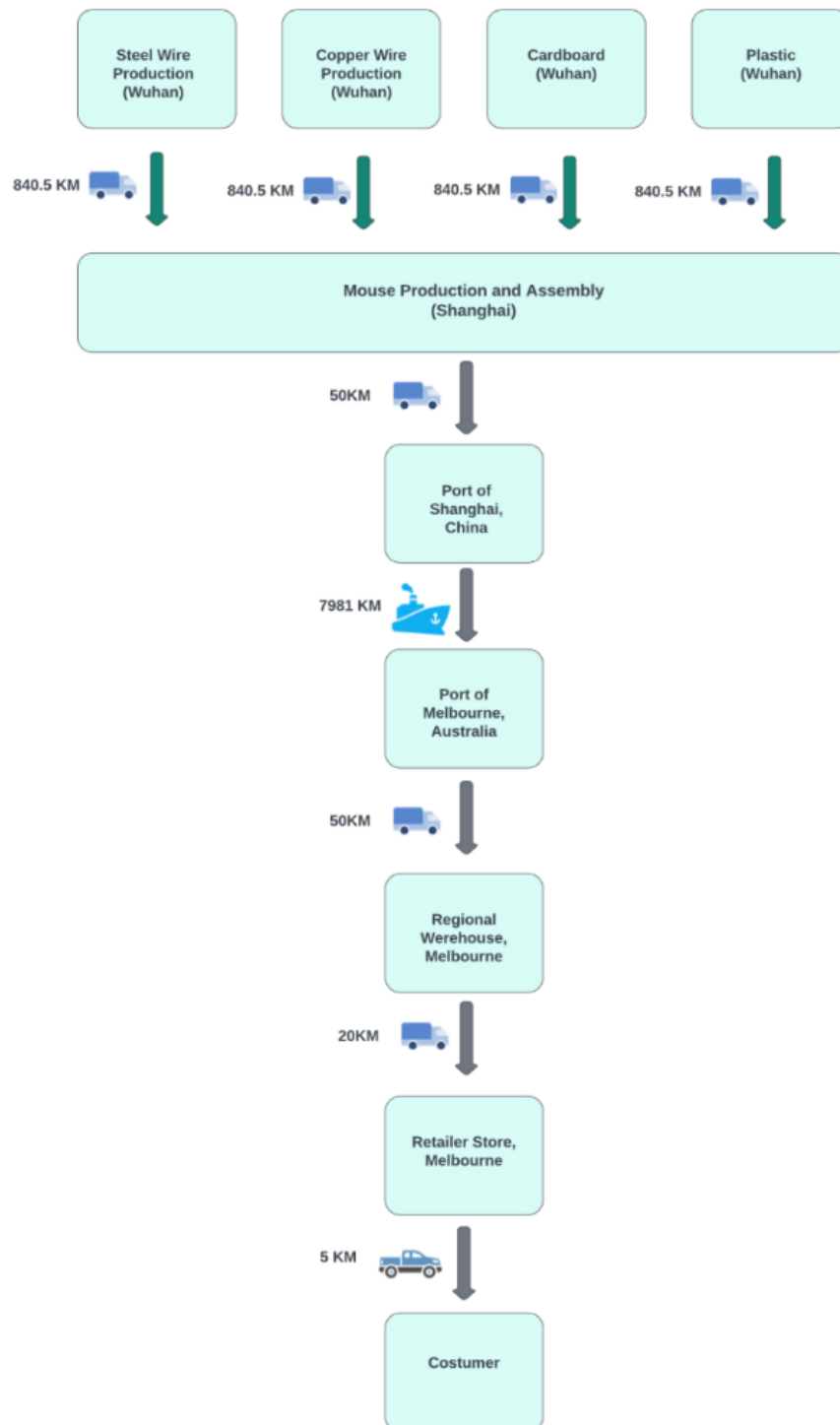


Fig. A.1 Visual representation of the transport distance assumptions

Table A.2 Midpoint impacts of the computer mouse

Name	Unit	Material	Manufacturing/ Assembly	Use	End of Life	Transport	Total
Fine particulate matter formation	kg PM2.5 eq	0.002429314	1.62E-04	0.006151	1.59E-05	4.18E-05	0.0088
Fossil resource scarcity	kg oil eq	0.314682482	0.024540489	0.134567	0.002634617	0.002632715	0.479058
Freshwater ecotoxicity	kg 1,4-DCB	0.318033446	0.001809279	0.313572	0.089633543	2.38E-04	0.723286
Freshwater eutrophication	kg P eq	8.58E-04	3.05E-05	5.01E-04	6.41E-06	1.51E-06	0.001397
Global warming	kg CO2 eq	1.123097038	0.094028742	0.486409	0.035508008	0.008984438	1.748027
Human carcinogenic toxicity	kg 1,4-DCB	0.099733788	0.004668955	0.237411	0.002760314	8.26E-04	0.3454
Human non-carcinogenic toxicity	kg 1,4-DCB	3.282256335	0.0332874	3.232637	0.395715347	0.004372899	6.948269
Ionizing radiation	kBq Co-60 eq	0.092010963	0.009182433	0.09177	2.16E-04	1.48E-04	0.193327
Land use	m2a crop eq	0.028816867	0.003306045	0.021676	4.33E-04	1.21E-04	0.054353
Marine ecotoxicity	kg 1,4-DCB	0.404473977	0.00243501	0.404153	0.112636139	3.29E-04	0.924028
Marine eutrophication	kg N eq	5.73E-05	2.35E-06	5.24E-05	1.30E-04	1.15E-07	0.000243
Mineral resource scarcity	kg Cu eq	4.39E-04	1.13E-05	0.002648	6.88E-07	5.17E-07	0.003099
Ozone formation, Human health	kg NOx eq	0.002675537	2.26E-04	0.002015	3.69E-05	1.27E-04	0.005081
Ozone formation, Terrestrial ecosystems	kg NOx eq	0.002746305	2.30E-04	0.002057	3.77E-05	1.29E-04	0.0052
Stratospheric ozone depletion	kg CFC11 eq	5.20E-07	3.03E-08	3.38E-07	1.54E-08	5.67E-09	9.1E-07
Terrestrial acidification	kg SO2 eq	0.004840274	3.22E-04	0.019902	2.86E-05	1.27E-04	0.02522
Terrestrial ecotoxicity	kg 1,4-DCB	19.13710354	0.056436653	25.3903	0.022297269	0.020905264	44.62704
Water consumption	m3	6.285950061	1.395030269	6.415947	0.014142444	0.007221363	14.11829

Table A.3 Environmental impacts from different transport modes

Name	Unit	China Train	China Road	India Road
Fine particulate matter formation	kg PM2.5 eq	4.18E-05	4.22E-05	4.53E-05
Fossil resource scarcity	kg oil eq	0.002632715	0.002822635	0.002855015
Freshwater ecotoxicity	kg 1,4-DCB	2.38E-04	2.46E-04	2.43E-04
Freshwater eutrophication	kg P eq	1.51E-06	1.53E-06	1.51E-06
Global warming	kg CO2 eq	0.008984438	0.009470969	0.009645677
Human carcinogenic toxicity	kg 1,4-DCB	8.26E-04	7.97E-04	8.16E-04
Human non-carcinogenic toxicity	kg 1,4-DCB	0.004372899	0.00475595	0.004564418
Ionizing radiation	kBq Co-60 eq	1.48E-04	1.55E-04	1.53E-04
Land use	m2a crop eq	1.21E-04	1.49E-04	1.30E-04
Marine ecotoxicity	kg 1,4-DCB	3.29E-04	3.49E-04	3.41E-04
Marine eutrophication	kg N eq	1.15E-07	1.16E-07	1.16E-07
Mineral resource scarcity	kg Cu eq	5.17E-07	5.28E-07	5.32E-07
Ozone formation, Human health	kg NOx eq	1.27E-04	1.29E-04	1.39E-04
Ozone formation, Terrestrial ecosystems	kg NOx eq	1.29E-04	1.31E-04	1.40E-04
Stratospheric ozone depletion	kg CFC11 eq	5.67E-09	5.91E-09	6.13E-09
Terrestrial acidification	kg SO2 eq	1.27E-04	1.28E-04	1.38E-04
Terrestrial ecotoxicity	kg 1,4-DCB	0.020905264	0.031217356	0.026911543
Water consumption	m3	0.007221363	0.007052923	0.00693439